

1. Towers of Hanoi

Constraint: Moves allowed only $A \leftrightarrow B$ and $B \leftrightarrow C$ (not $A \leftrightarrow C$).

$M(n)$ = moves for $A \rightarrow C$ (non-adjacent)

$M'(n)$ = moves for $A \rightarrow B$ (adjacent)

Algorithm for $A \rightarrow C$:

- | | |
|--|--------------|
| 1. Move $(n - 1)$ disks: $A \rightarrow C$ | $[M(n - 1)]$ |
| 2. Move disk n : $A \rightarrow B$ | $[1]$ |
| 3. Move $(n - 1)$ disks: $C \rightarrow A$ | $[M(n - 1)]$ |
| 4. Move disk n : $B \rightarrow C$ | $[1]$ |
| 5. Move $(n - 1)$ disks: $A \rightarrow C$ | $[M(n - 1)]$ |

$$M(n) = 3M(n - 1) + 2, \quad M(1) = 2$$

$$M(n) = 3^n - 1$$

$$M(1) = 2 \implies 3A - 1 = 2 \implies A = 1$$

$M(n) = 3^n - 1$

2. Partition

(a) **begin**
 PARTITION($A[1..n], p$)
 $k \leftarrow 0$
 for $i \leftarrow 1$ **to** n **do**
 if $A[i] \leq p$ **then then**
 $k \leftarrow k + 1$
 SWAP($A[k], A[i]$)
 return k

Start of each loop iteration:

$$A[1..k] \leq p$$

$$A[k + 1..i - 1] > p$$

When the algorithm ends:

$$A[1..k] \leq p$$

$$A[k + 1..n] > p$$

(b)

Comparisons: n

Swaps: $\leq n$

$T(n) = O(n)$

3. Selection Recurrence

(a) Groups of size 3:

$$\begin{aligned}
 \text{Groups:} & \quad n/3 \\
 \text{Elements} \leq \text{group median:} & \quad 2 \\
 \text{Medians} \leq \text{pivot:} & \quad \geq \frac{1}{2} \cdot \frac{n}{3} = \frac{n}{6} \\
 \text{Guaranteed} \leq \text{pivot:} & \quad 2 \cdot \frac{n}{6} = \frac{n}{3} \\
 \text{Worst-case recurse:} & \quad n - \frac{n}{3} = \frac{2n}{3}
 \end{aligned}$$

$$T(n) = T\left(\frac{n}{3}\right) + T\left(\frac{2n}{3}\right) + O(n)$$

Note: $\frac{1}{3} + \frac{2}{3} = 1 \implies T(n) = \Theta(n \log n)$

(b) Groups of size $2k + 1$:

$$\begin{aligned}
 \text{Groups:} & \quad \frac{n}{2k+1} \\
 \text{Elements} \leq \text{group median:} & \quad k+1 \\
 \text{Medians} \leq \text{pivot:} & \quad \geq \frac{n}{2(2k+1)} \\
 \text{Guaranteed} \leq \text{pivot:} & \quad (k+1) \cdot \frac{n}{2(2k+1)} = \frac{(k+1)n}{2(2k+1)} \\
 \text{Worst-case:} & \quad n - \frac{(k+1)n}{2(2k+1)} = n \cdot \frac{2(2k+1) - (k+1)}{2(2k+1)} \\
 & \quad = n \cdot \frac{3k+1}{2(2k+1)}
 \end{aligned}$$

$$T(n) = T\left(\frac{n}{2k+1}\right) + T\left(\frac{(3k+1)n}{2(2k+1)}\right) + O(n)$$

Linear time condition:

$$\frac{1}{2k+1} + \frac{3k+1}{2(2k+1)} = \frac{3(k+1)}{2(2k+1)} < 1 \iff k > 1$$

4. Intersection Detection

begin

INTERSECTIONDETECT($I[1..n]$)

Sort I by $A[i]$

$m \leftarrow B[1]$

for $i \leftarrow 2$ **to** n **do**

if $A[i] < m$ **then**

return true

$m \leftarrow \max(m, B[i])$

return false

$\forall j < i : A[j] \leq A[i]$

overlap $\iff A[i] < B[j]$

$T(n) = 2T(n/2) + O(n)$

Sorting: $O(n \log n)$

Scan: $O(n)$

$$T(n) = O(n \log n)$$